ARNAV ATUL WAGHDHARE

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EDUCATION

- **KJ Somaiya College of Engineering** *August 2023 - Present*
  B.Tech in Computer Engineering Mumbai, India
  - CGPA: 8.84

EXPERIENCE

- **Full Stack Developer** *May 2026 - August 2026*
  Datafuel Mumbai, India
  - Engineered end-to-end **web scraping & ETL pipelines** by reverse engineering private APIs, and developing **ReactJS** analytics dashboards for enterprise clients for asynchronous task management.
  - Built high-throughput Python data pipelines using **multithreading/multiprocessing** to scrape **1,000+ darkstores** in 4 hours, generating **15M+ records** in ClickHouse while bypassing anti-bot systems.
- **Associate Mentor** *July 2024 - April 2025*
  Somaiya Machine Learning Research Association Mumbai, India
  - Delivered a hands-on ML workshop for **20+** students.
  - Explained key concepts like **Support Vector Machines (SVM)** and **K-Nearest Neighbors (KNN)**.
  - Leveraged engaging presentation skills to break down complex concepts, fostering enthusiastic participation and clearer ML understanding among attendees.

PUBLICATIONS

C=CONFERENCE, P=IN PUBLISHING

[C.1] **A. Waghdhare**, A. Bhowmik, I. Bhoyar, S. Sankhe and K. Kulkarni, et al. (2026). **Smart Farming Innovations for Maharashtra Using ML with Dual-Node LoRaWAN Connectivity**. In *IEEE International Conference for Convergence in Computing Technology (I3CTCON)*, pp. 1-9. 2026, Lonavala, India. DOI: 10.1109/I3CTCON68242.2026.11507500

[P.1] **A. Waghdhare**, P. Mestry, S. Sankhe, et al. (2026). **Gesture-Based Assistive Tool for Children with Speech and Hearing Impairments**. Developed Vision Models trained on Indian Sign Language. Presented in *6th International Conference on Computer Vision and Robotics (CVR 2026)*, publishing in Springer 2026.

PROJECTS

- **Astra: Low-Latency Streaming VAD & Wake-Word Pipeline for Voice Assistants** *July 2026*
  Tools: WebSockets, PyTorch, ONNX, Python, Go [GitHub]
  - Engineered an end-to-end real-time streaming speech pipeline for conversational AI. Developed a and integrated Custom VAD, wake-word spotting, utterance endpointing, Whisper ASR, and streamed LLM responses. Optimized model latency and CPU inference throughput via ONNX Runtime.
  - Trained and deployed custom VAD and OpenWakeWord models (PyTorch → ONNX Runtime), 96.4% VAD recall and 96% wake-word recall at <1 false-accept/hour, serving CPU-only in-process inference at 45 µs/frame (450× real-time).
- **Amazon ML Challenge: Multimodal Price Prediction System** *November 2025*
  Tools: PyTorch, Hugging Face Transformers (CLIP), FAISS, HDF5 [GitHub]
  - Built a multimodal model fusing frozen **CLIP** vision-language embeddings through a **Transformer encoder**, scoring a top-ranked 77% SMAPE Score on a 20–25GB, 75K-product dataset for the Amazon ML Challenge 2025.
  - Applied parameter-efficient transfer learning (1.7M/153M params trained), cutting validation loss 24% via AdamW + cosine-annealing scheduling, and benchmarked it against a **CLIP+FAISS KNN gradient-boosting** baseline.

SKILLS

- **Programming Languages:** Python, C++, Java, GoLang, JavaScript, SQL, Kotlin, Shell, HTML, CSS
- **Deep Learning & Machine Learning:** PyTorch, TensorFlow, Hugging Face Transformers, ONNX Runtime, OpenCV, Scikit-learn, NumPy, Pandas
- **LLMs & Generative AI:** LangChain, Google ADK, Ollama, OpenRouter, RAG, Prompt Engineering, Fine-tuning, Vector Databases
- **Backend & Infrastructure:** FastAPI, REST APIs, WebSockets, Docker, Celery, ClickHouse, PostgreSQL, Git

ACHIEVEMENTS & CERTIFICATIONS

- **DeepLearning.ai:** Supervised Machine Learning: Regression and Classification *November 2024*
- **DeepLearning.ai:** Advanced Learning Algorithms *December 2024*
- **IUCEE:** Mastering Multi-Agent AI: Building Intelligent Collaborative System *December 2024*
- **GAJSHIELD 2025:** Built a model that ranked my team in the Top 5 in GAJSHIELD Hackathon *April 2025*